



RESEARCH COMIC

A FAIR Comic about Research Data Infrastructure

Part 1: From Charlemagne to ing.grid

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Date Submitted:

2026-05-30

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Keywords:

Comic, Inggid

Data availability:

Data can be found here: <http://doi.org/10.5281/zenodo.19468332>

Software availability:

-

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Abstract. Comics journalism constitutes a graphic form of literary journalism. According to John S. Bak and Christopher Craig [1], comics journalism and literary journalism share key features: they both read like fiction but are based on immersive research, eyewitness accounts, and verifiable facts scrutinized as closely as possible. RWTH Aachen University has recently embraced comics journalism as a means of communicating ongoing research at the Chair of AI Methodology within the Faculty of Computer Science [2]. To enable close observation and immersive documentation of scientific practice, comic artist Alfred Neuwald was invited to work alongside the researchers. The resulting collaboration established the general framework for this project. Here we present the first result of a modified artist-in-residence concept implemented by the German National Research Data Infrastructure for Engineering Sciences (NFDI4ING).

1 **Jump to:** [Article](#) | [English comic](#) | [German comic](#)

2 1 Introduction

3 Inviting an outsider to observe and take part in scientific practice has a methodological history that
4 can be traced back to the ethnographic approach of Bruno Latour and Steve Woolgar's *Laboratory*
5 *Life* [3], [4]. *Laboratory Life* is a foundational work in science and technology studies. The book
6 includes a floor plan of the research laboratories, offices, and meeting rooms in which Latour
7 worked alongside the researchers. It also reproduces photographs of the working environment
8 and, in some cases, of the researchers. Latour and Woolgar thus document interactions with
9 and among the researchers in a spatially situated manner. They report on the verbal content of
10 these interactions and the vocabulary used. Their main focus is the researchers' investment in
11 the iterative production process that brings about the principal outcome of scientific work: peer-
12 reviewed publications. Crucially, Latour and Woolgar also provide transcripts and occasionally
13 accompany them with in-line commentary. The in-line commentary is used to describe aspects
14 of spatially situated interaction that are not captured in the transcripts themselves, including
15 deictic gestures—gestures that establish spatial reference within a shared environment. A key
16 constraint of in-line commentary, however, is that it competes for space with the transcript
17 itself: as descriptions of embodied and spatially situated action become more detailed, the

transcript becomes increasingly fragmented and more difficult to read. To address this problem, multimodal forms of transcription typically move such commentary into a separate column at the page margins (see, for example, [5]). This layout allows verbal interaction and descriptions of embodied action to be presented simultaneously without interrupting the flow of the transcript. Moreover, this layout permits the integration of graphical representations alongside the transcript, which constitutes a comparatively recent development in conversation analytic research (see, for example, [6]).

Comics represent embodied speech. They do so by depicting speakers and listeners within their environment together with spoken utterances, deictic gestures, and, in some cases, internal thoughts represented through thought balloons. As a graphic medium, comics are therefore particularly well suited to documenting multimodal interaction in situated research environments. Comics, however, usually transport readers into the realm of imagination rather than documented fact. Comics journalism—together with its European and Japanese counterparts, *BD reportage* and documentary manga—differs precisely in this respect [1], [7].

At RWTH Aachen University, comics journalism has recently been embraced as a means of communicating ongoing research at the Chair of AI Methodology within the Faculty of Computer Science [2]. The collaboration between researchers and comic artist Alfred Neuwald established the general framework for the present project. By combining ethnographic observation with the multimodal representational capacities of comics, we explore comics journalism as a method for documenting situated research practices within the administrative office of the German National Research Data Infrastructure for Engineering Sciences (NFDI4ING).

2 Result

2.1 A FAIR Comic on Research Data Infrastructure

Dear Reviewers, please note: The line numbers in the following pages refer to comic panels, not to individual text lines.

From Charlemagne to Little Charles.
 From Carolingian minuscule via Middle Francia to  NFDI4ING.*
 Research data infrastructure is and remains an exciting subject.



* NFDI4ING is the consortium for engineering sciences within the German National Research Data Infrastructure (NFDI)

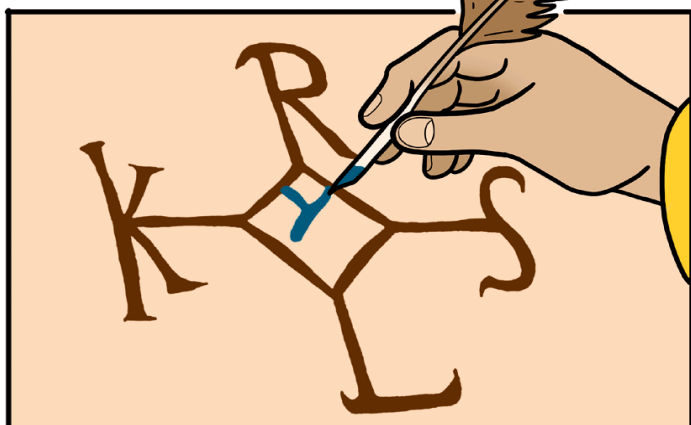
44

In the 8th/9th century, Charlemagne began building a sustainable knowledge infrastructure through his centralized culture of writing. He relied on a network of scribes in monasteries and at his court in Aachen.



45

Charlemagne could not write. When signing documents, he merely added the so-called completion stroke to the Karolus monogram.



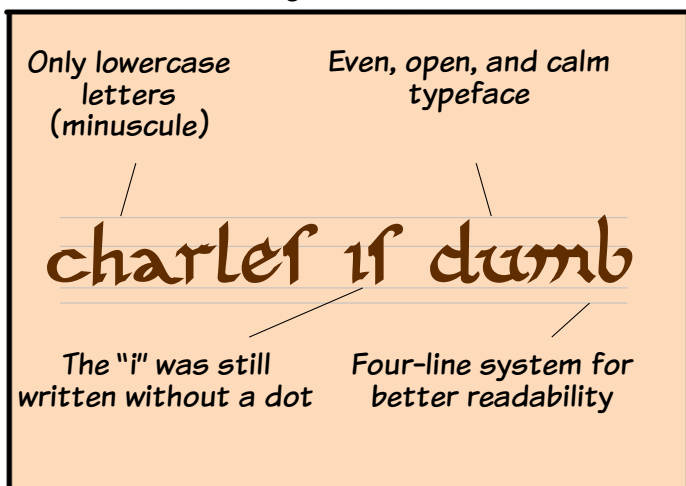
However, Charlemagne promoted education by having scholars dictate, copy, and distribute documents.

46

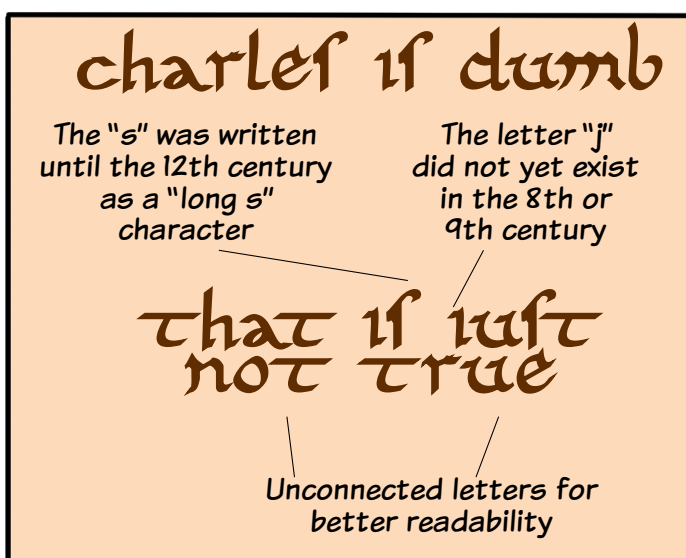


47

To ensure the readability and consistency of documents throughout his vast empire, he commissioned the development of the "Carolingian minuscule".

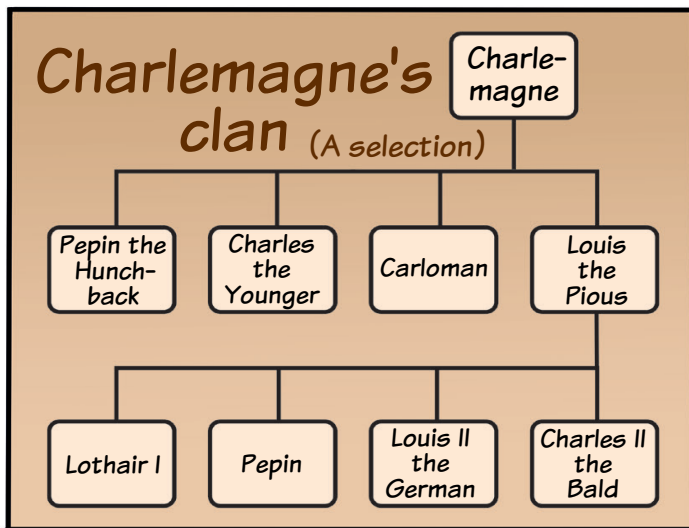


48



49

Charlemagne's sons and grandsons continued this analogue research infrastructure based on parchment and ink.



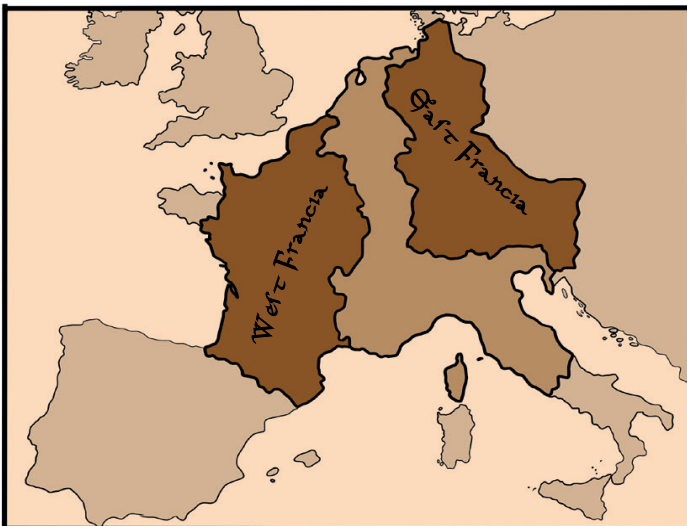
50

One of his grandsons, Lothair I, played a major role in the Treaty of Verdun in 843.



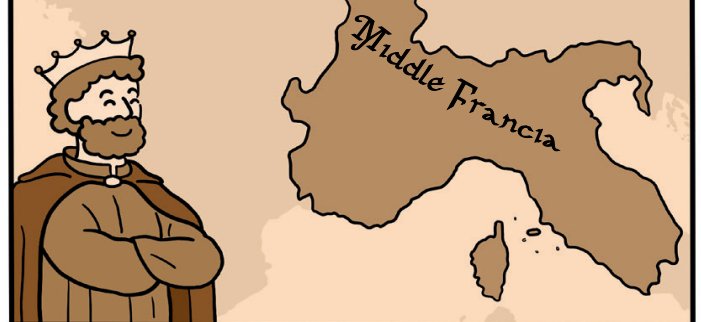
51

This treaty is regarded as the birth of the predecessors of France and Germany (West Francia and East Francia).



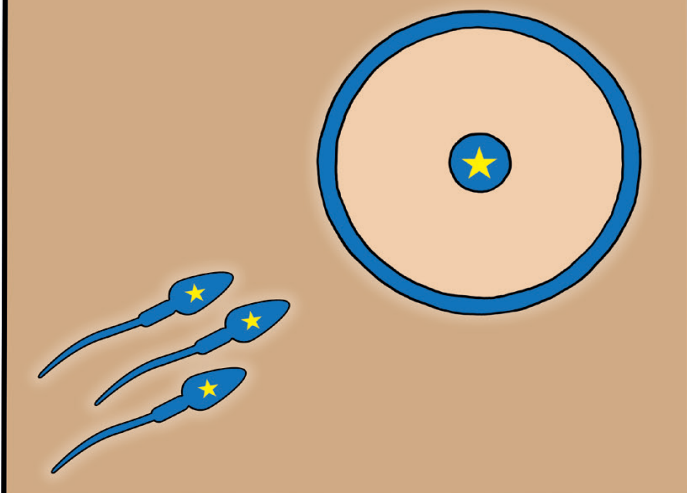
52

Lothair himself received Middle Francia. It stretched as a long, narrow corridor from the North Sea to the Mediterranean.



53

This Middle Francia became the nucleus of European integration.



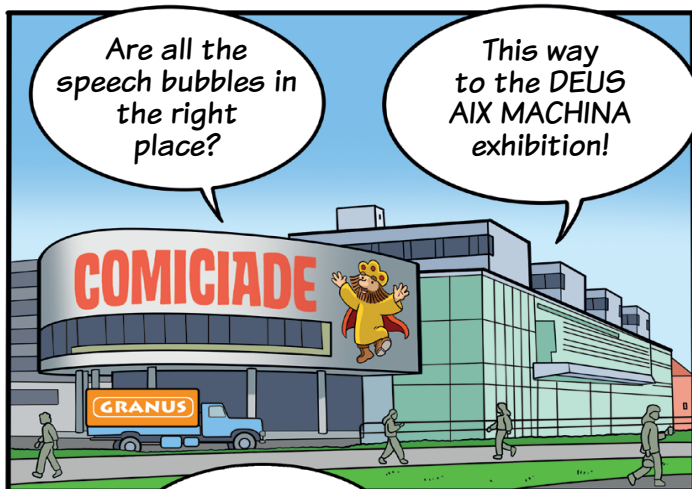
54

Many institutions of the European Union are now located along this former corridor (Brussels, Luxembourg, Strasbourg, Rome).



55

And that is how the Comiciade at the Technology Centre on Europaplatz in Aachen opens its doors to comic fans from all over the world.

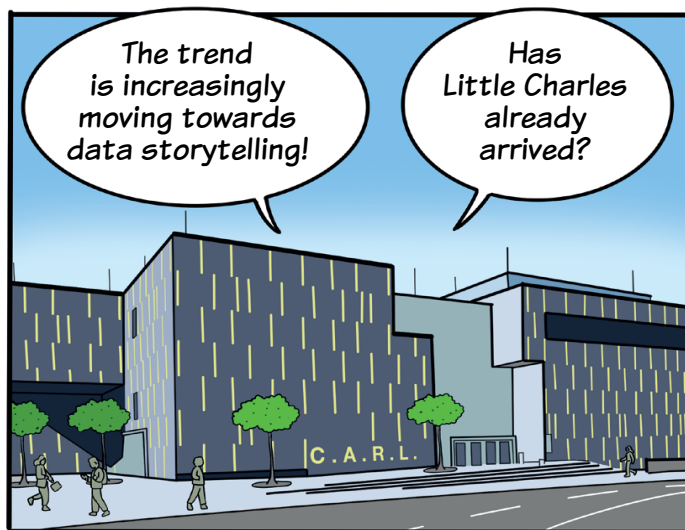


57



56

Preparations are already under way at RWTH Aachen's C.A.R.L. as well.



58



59



60



61



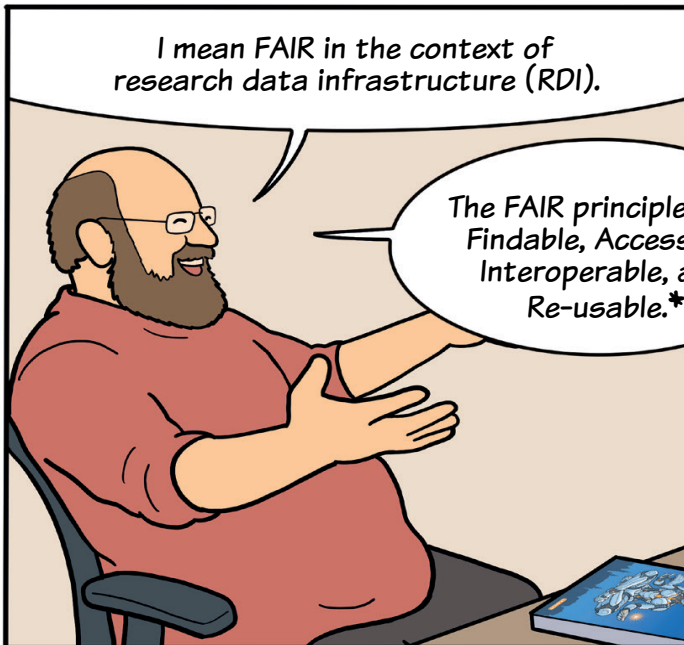
I really enjoyed your comic about your time at the Chair of Methodology of Artificial Intelligence. What do you think about making the world's first FAIR comic now?

62



The world's first FAIR comic ever?

63



I mean FAIR in the context of research data infrastructure (RDI).

The FAIR principles are: Findable, Accessible, Interoperable, and Re-usable.*

64



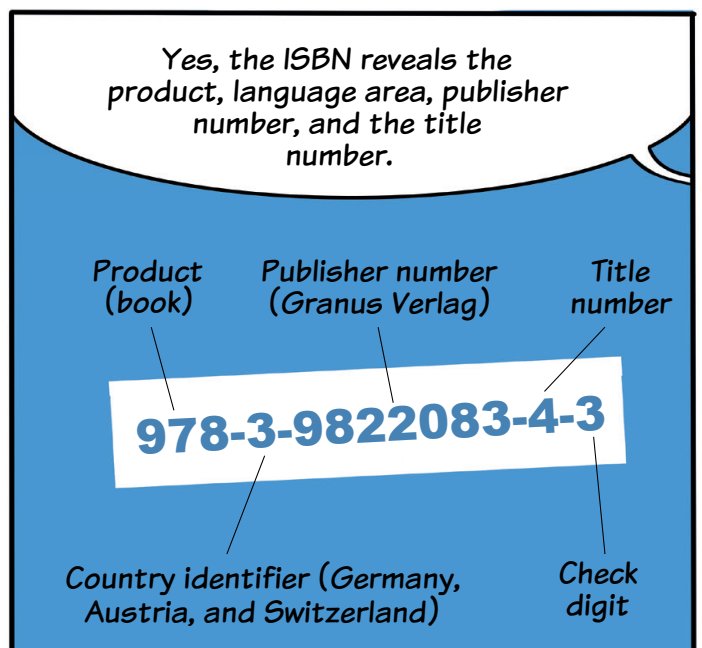
You'll have to explain that to me, Évariste.

65



It's quite simple. On the back of your comic, you included an ISBN so it can be found more easily in bookshops.

66



*Findable, accessible, interoperable, and re-usable

67

You need an ISBN to be included in the German Books in Print directory (VLB). If you are not listed there, it is very difficult to get your book into shops.



68

The VLB is the central, independent database and research tool for the German-language book trade.



69

Interesting. We researchers use a DOI (Digital Object Identifier) instead of an ISBN.



70

It is a unique, persistent identifier for objects of all kinds. An object may have several DOIs, but a DOI always points to only one object.

DOI resolver

Repository

<https://doi.org/10.5281/zenodo.19468332>

Prefix

Or as a QR code



Object

71

A partner of NFDI4ING, the DataCite association, assigns DOIs.*



72

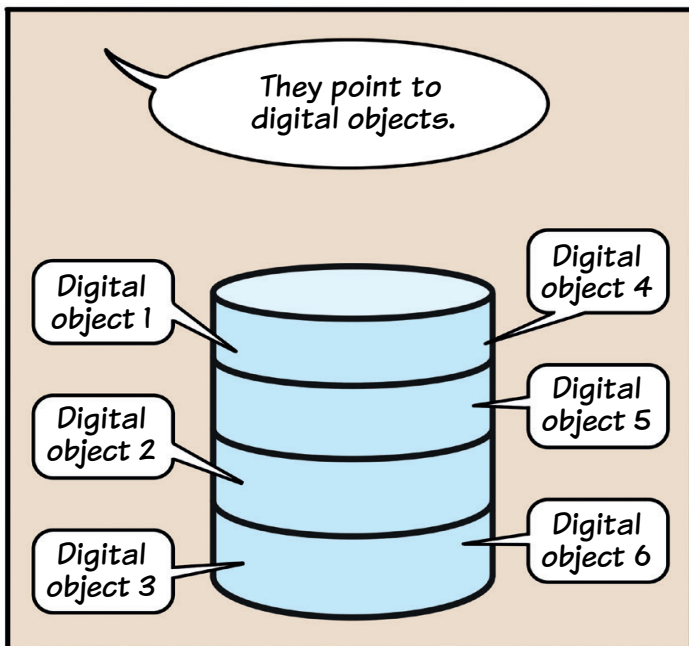
DOIs are the standard in academic publications for precise source references.

And how exactly do they work?



*DataCite is a registration agency for DOIs

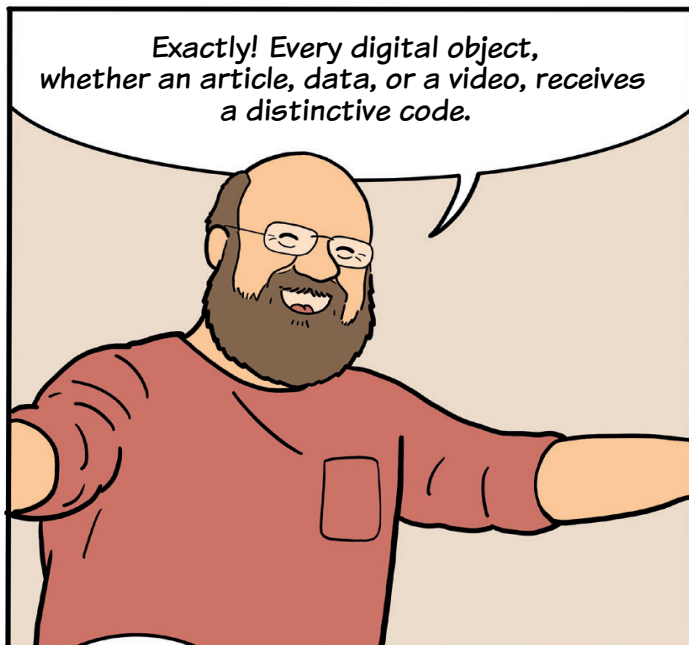
73



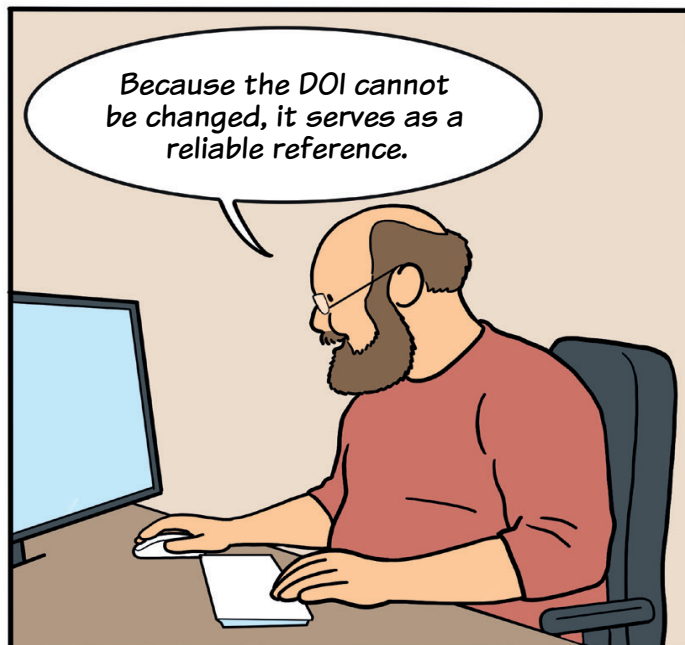
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76



77



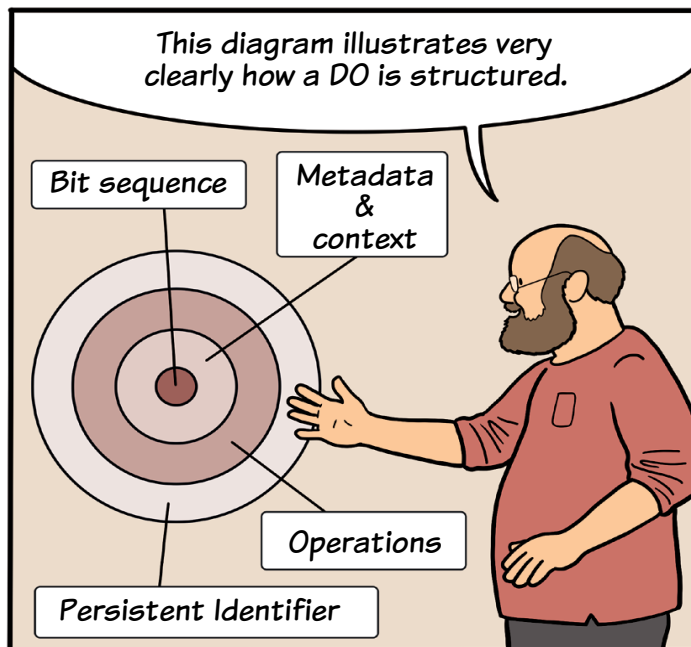
78



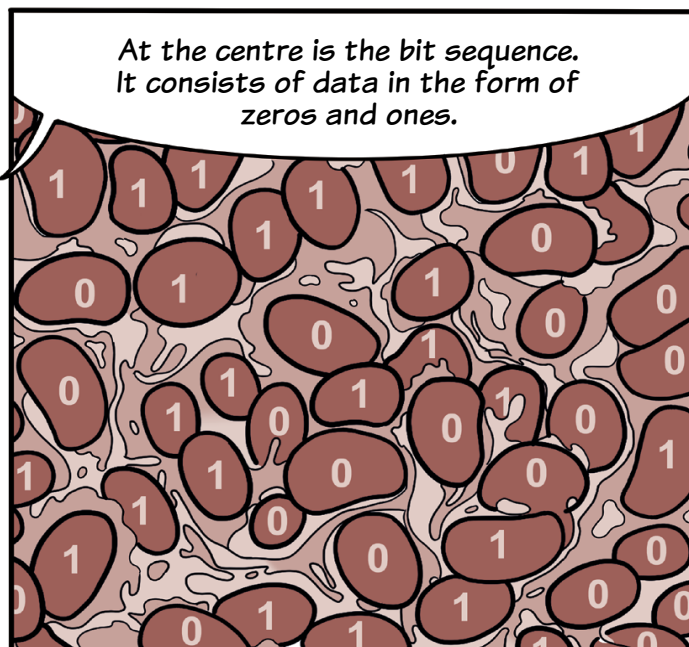
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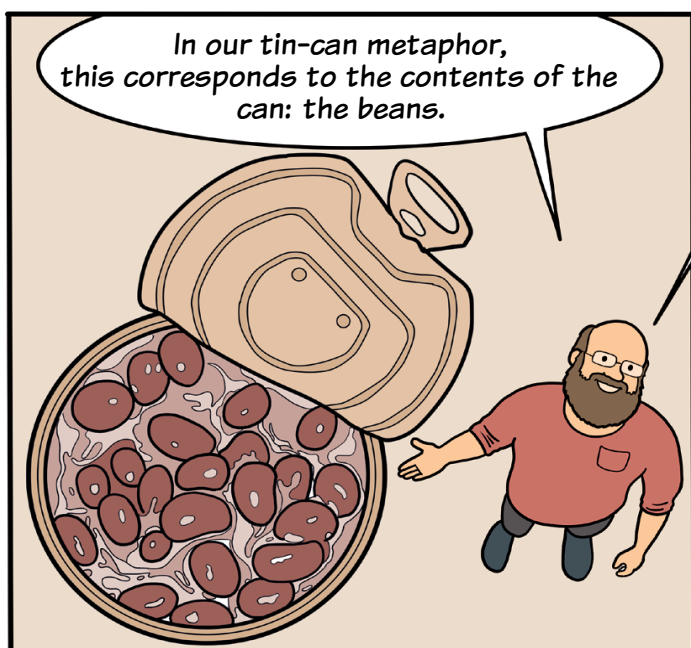
80



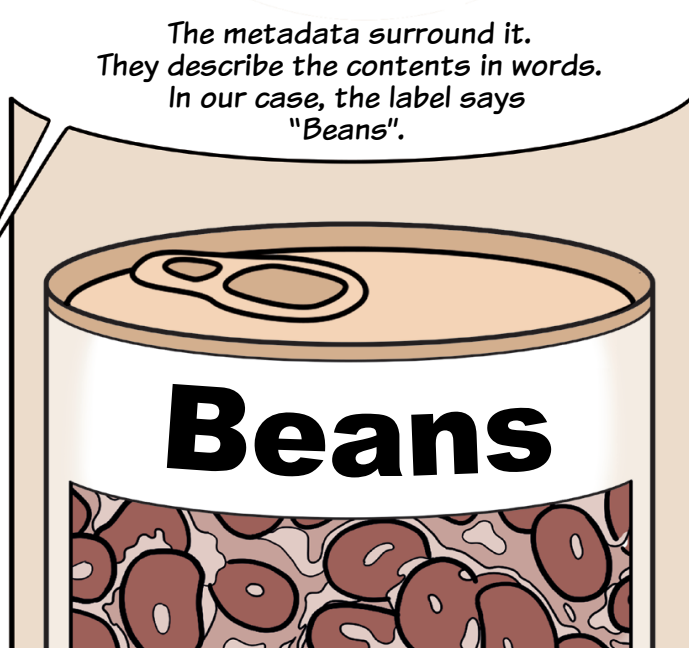
81



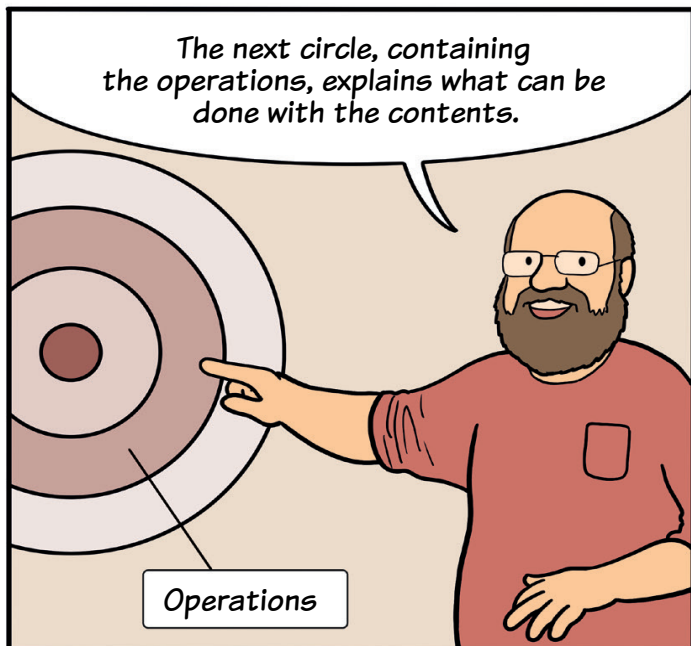
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83



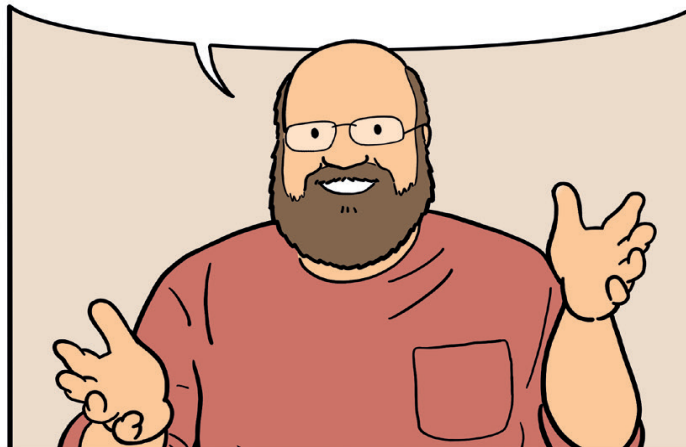
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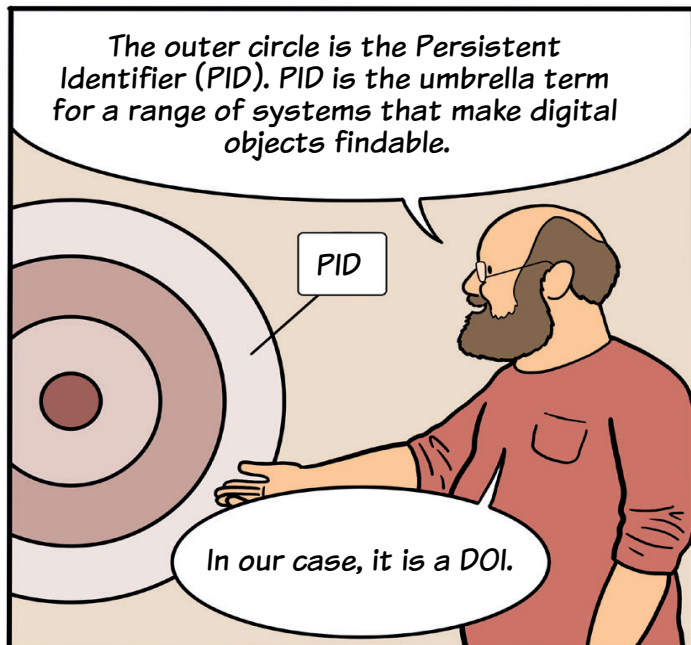
85



NFDI4ING is working to define more than just a download function within the operations.



86

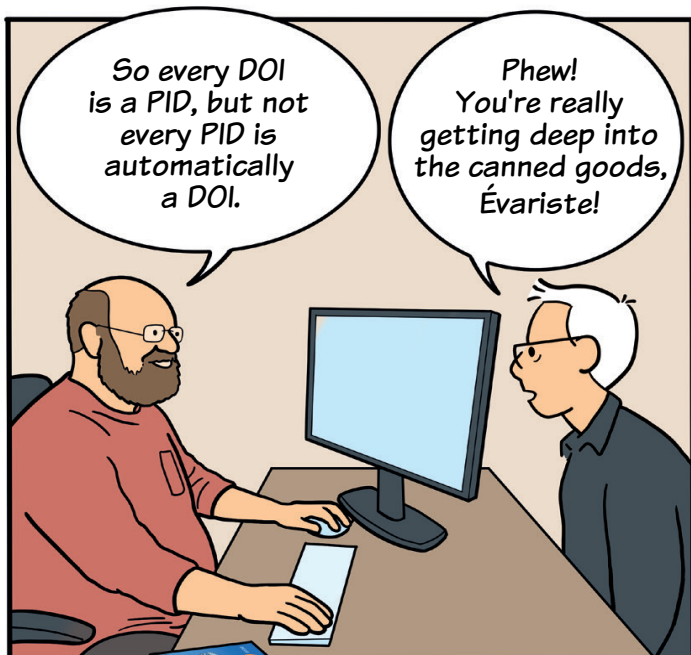


87

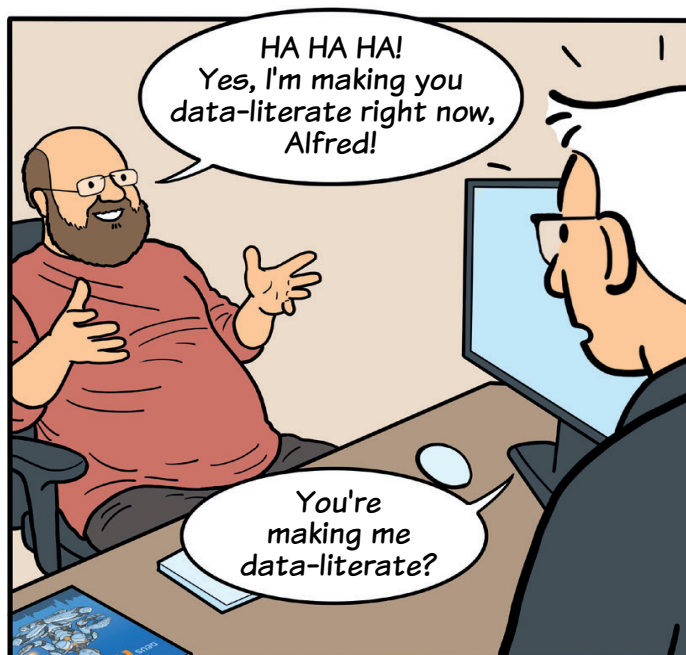
However, there are also other systems, such as URN, Handle, or ORCID.*



88



89



*URN = Uniform Resource Name, Handle = Handle System; ORCID = Open Researcher and Contributor ID

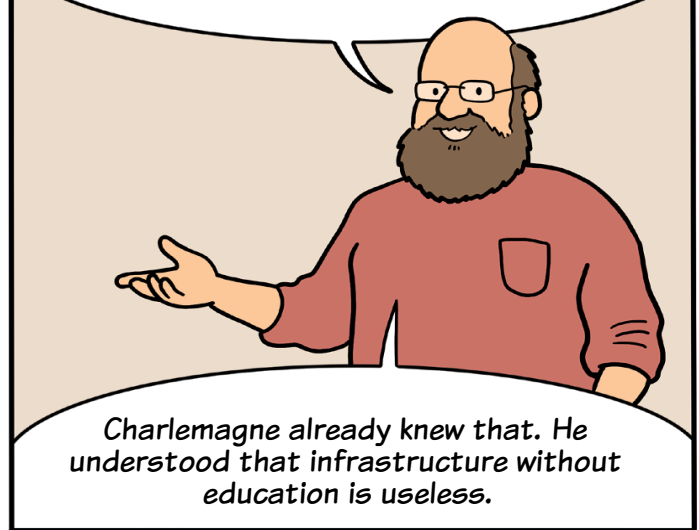
90

In RDI, the term "data literacy" describes the process of acquiring the fundamental skills and knowledge required to read and write research data.



91

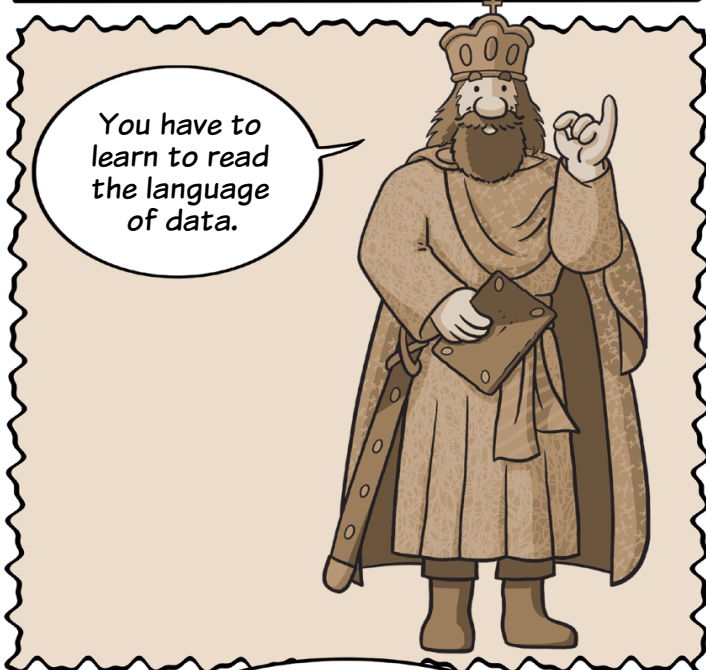
If nobody can read them, even the best archive is useless, and the data remain silent.



Charlemagne already knew that. He understood that infrastructure without education is useless.

92

You have to learn to read the language of data.



93

Charlemagne placed great importance on clerics and administrative officials being able to read and write — also so that they could copy texts correctly. Faulty data transmission was already a problem back then.

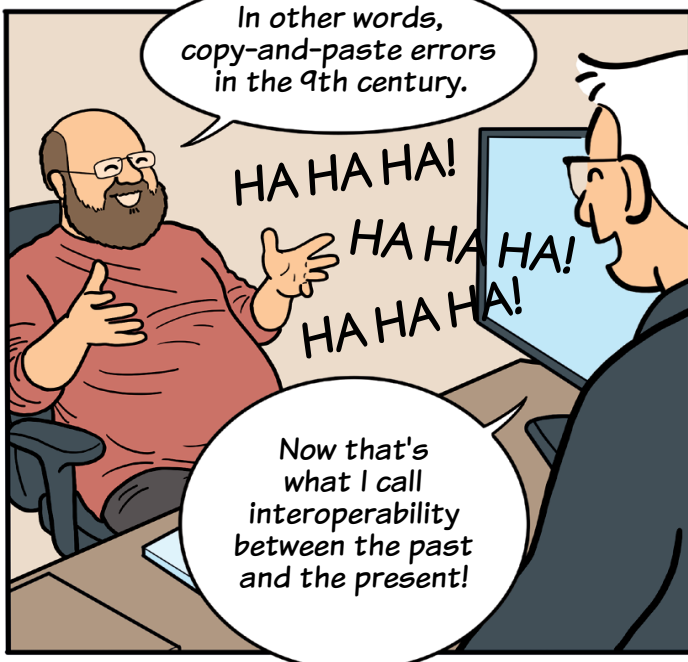


94

In other words, copy-and-paste errors in the 9th century.

HA HA HA!
HA HA HA!
HA HA HA!

Now that's what I call interoperability between the past and the present!



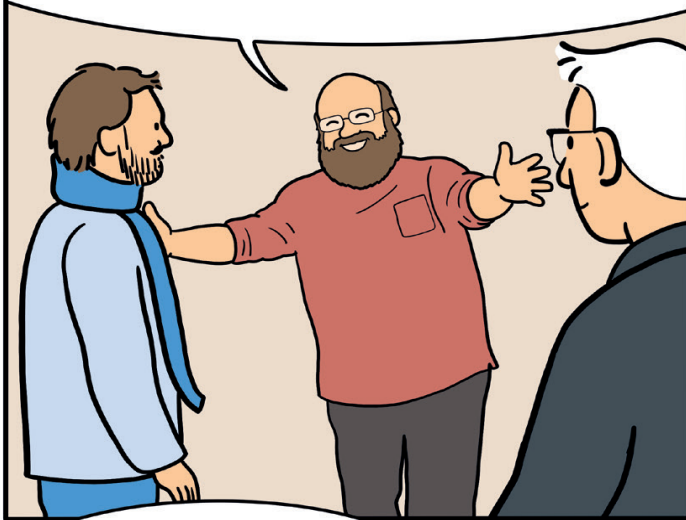
95

What are you laughing about?



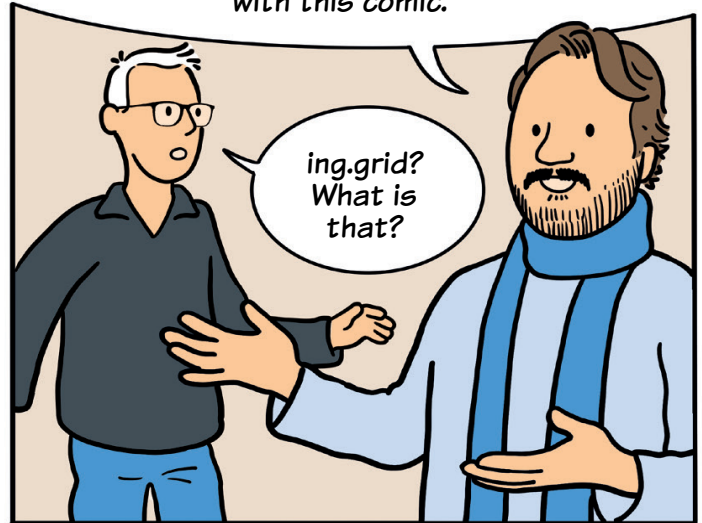
96

Hello, Tobias. We are currently creating a FAIR comic, and it is proving very entertaining.



97

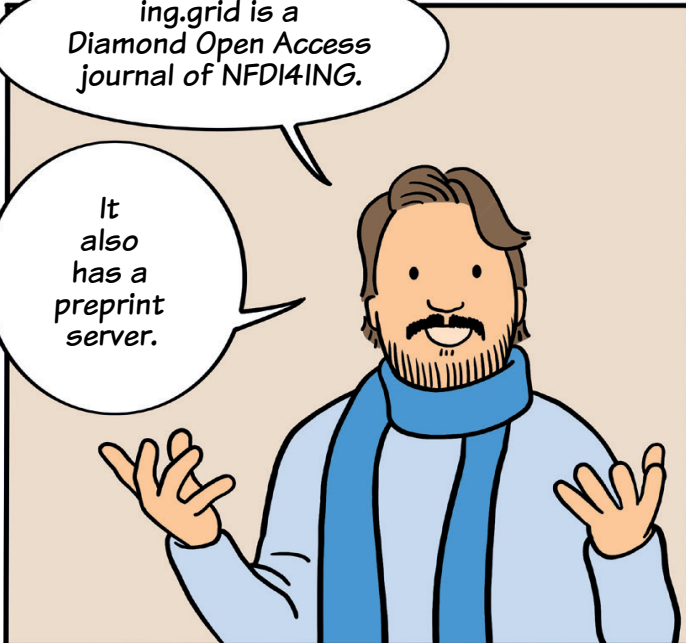
That sounds exciting. Yesterday, I submitted my latest paper to ing.grid. You could do the same with this comic.



98

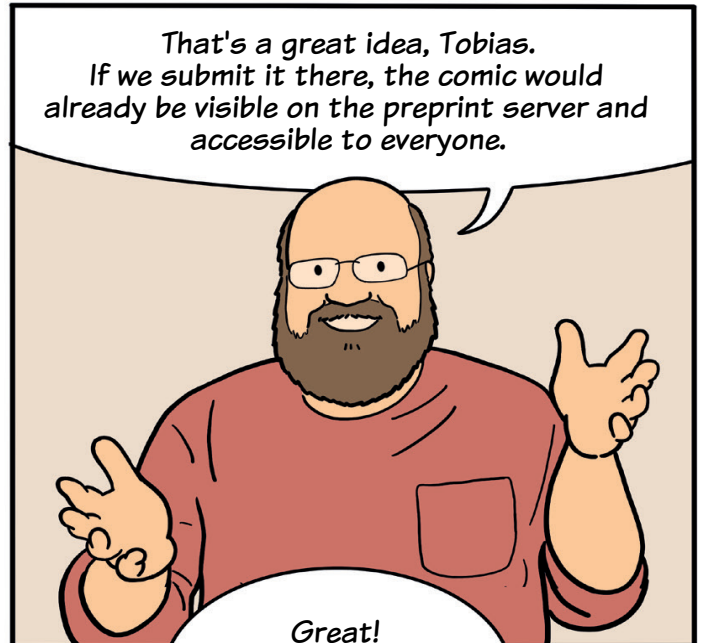
ing.grid is a Diamond Open Access journal of NFDI4ING.

It also has a preprint server.



99

That's a great idea, Tobias. If we submit it there, the comic would already be visible on the preprint server and accessible to everyone.



100

Exactly! A community review would be the first step there. That means the comic could already be commented on and evaluated.



101

Great! Where should I put my completion stroke?



102

Just a moment!
First, we have to translate the comic into
the target language. And there is already
a guide for that.

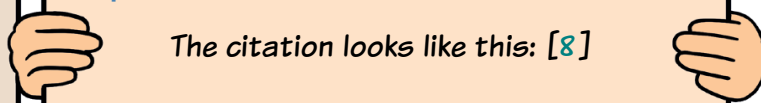


103

This guide can be found
under this HANDLE.

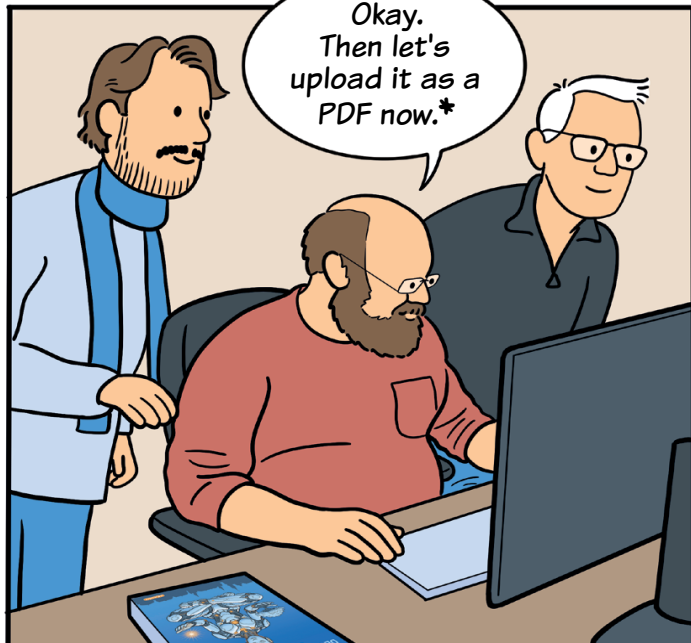
<http://hdl.handle.net/2268.2/12918>

The citation looks like this: [8]



104

Okay.
Then let's
upload it as a
PDF now.*



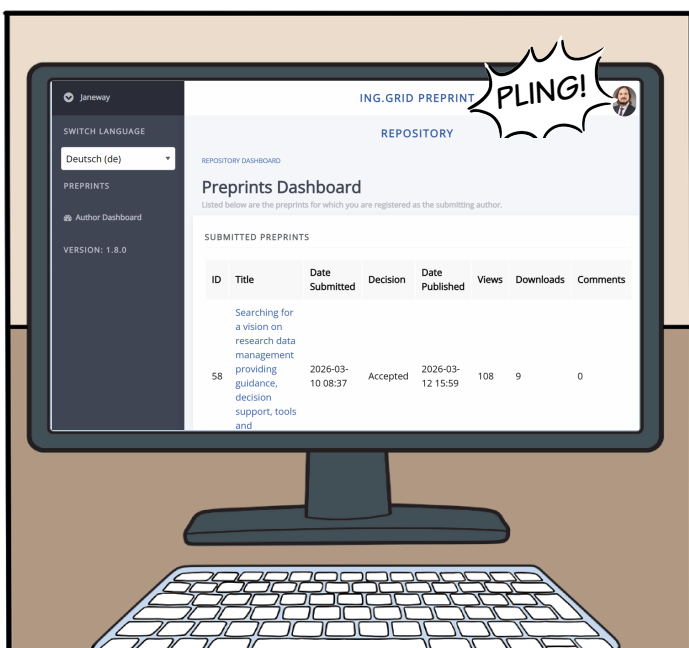
105

Aaaand...
Here we go!



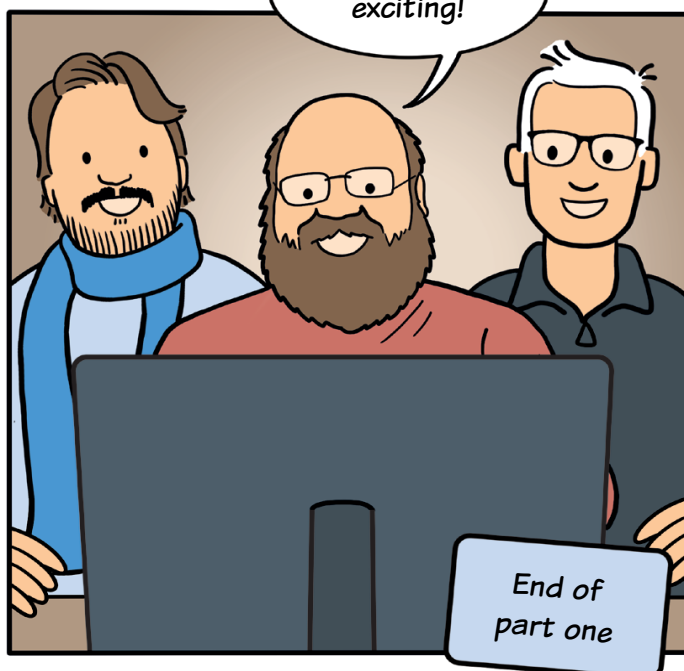
Now it gets
exciting!

106



107

End of
part one



* PDF = Portable Document Format

107 3 Materials and Method

108 3.1 Project Development and Collaboration

109 Évariste Demandt first contacted Alfred Neuwald in July 2025 and met with him shortly thereafter.
 110 During this initial meeting, he learned about Alfred’s ongoing collaboration with the Chair of AI
 111 Methodology. As this collaboration was expected to continue for several more months, Alfred’s
 112 availability for additional commitments was limited. Nevertheless, both parties agreed to remain
 113 in contact and continue exchanging ideas. Évariste suggested that Alfred might participate as a
 114 guest observer in the 2025 Conference on Research Data Infrastructure (CoRDI 2025), and Tobias
 115 Hamann subsequently extended a formal invitation. CoRDI 2025 took place at RWTH Aachen
 116 University from 26 to 28 August 2025 [9]. Alfred participated in the conference, including the
 117 evening event at the Laboratory for Machine Tools and Production Engineering (WZL). He used
 118 the opportunity to familiarize himself with the field of research data infrastructure and with the
 119 people involved.

120 Following the conference, Évariste convened a meeting between Alfred and representatives
 121 of NFDI4ING, IT Center and WZL, as well as a representative of the Chair of Society and
 122 Technology at RWTH Aachen University. It quickly became apparent that we lacked the resources
 123 to replicate Alfred’s ongoing collaboration with the Chair of AI Methodology, a project that
 124 enabled Alfred to work full-time with the researchers. As a result, we began exploring alternative
 125 formats for collaboration. Recognizing Alfred’s ability to ask questions, Évariste concluded that
 126 a format centered on regular conversations would best accommodate the budget constraints. A
 127 weekly *jour fixe* was therefore scheduled, during which Alfred visited NFDI4ING’s administrative
 128 office at the IT Center.

129 Alfred began work on the FAIR comic in February 2026. He photographed Évariste and, later,
 130 Tobias as reference material for the comic, which he created outside RWTH Aachen University
 131 at his atelier. The collaboration followed a meeting-only format: no emails, phone calls, or
 132 messages were exchanged between meetings in order to keep the workload manageable. This
 133 meant that Alfred always came in well prepared with questions and, later, drafts. Meetings were
 134 open-ended and typically exceeded one hour. Alfred’s questions sometimes revealed gaps in
 135 shared understanding of research data management (RDM) or exposed issues that could not be
 136 resolved immediately. In such instances, the questions were adjourned and additional expertise
 137 was sought from specialists. This occurred twice during the project: once with support from
 138 NFDI4ING’s Task Area Federated Interactive Data Space and once with support from the Chief
 139 Technical Officer of the IT Center.

140 During the weekly meetings, learning was reciprocal. Alfred shared his extensive knowledge of
 141 comic culture, referring to documentary works such as Art Spiegelman’s *Maus* [10], Marjane
 142 Satrapi’s *Persepolis* [11], and Guy Delisle’s *Pyongyang* [12]. Delisle’s work, in particular,
 143 received special attention as an outstanding example of observer commentary. Alfred admired
 144 Delisle’s ability to document a social world and occasionally drew parallels between his own
 145 encounters with researchers and Delisle’s experiences working in North Korea.

146 3.2 Documentary Basis and Narrative Framing

147 This is a work of literary journalism, which means that it is grounded in Alfred's research. The
 148 FAIR comic conveys his first-hand observations, conversations, and photographic documentation
 149 gathered during the weekly meetings at the administrative office of NFDI4ING at the IT Center
 150 of RWTH Aachen University. Among the materials that informed Alfred's work was a leaflet
 151 published by the Landesinitiative NFDI der Digitalen Hochschule NRW and the AG FDM
 152 Awareness [13]. Copies were available in the administrative office, and the leaflet became a
 153 recurring point of reference in conversations between Évariste and Alfred and, later, between
 154 Évariste, Tobias, and Alfred. The influence on the FAIR comic can be seen both in the range
 155 of subjects covered and in the canned-food metaphor, which originates on page 12 of the
 156 leaflet. Another material used by Alfred was a slide desk from NFDI4ING's Task Area Federated
 157 Interactive Data Space with a schematic representation of the different layers that constitute a
 158 FAIR Digital Object.

159 A point of contention was whether fictional elements should be included at all. At the outset of
 160 the project, Évariste was unfamiliar with the distinction between fictional comics and comics
 161 journalism and proposed narrative devices drawn from fictional comics. Alfred generally resisted
 162 such proposals. The need to reconcile them with the request to begin the story in 843 AD
 163 ultimately led to the inclusion of a few fictional elements on the first three pages, which are
 164 examined below. Before turning to these elements, however, we first explain Évariste's request
 165 to begin the story in 843 AD.

166 At the time of writing, Aachen has presented the International Charlemagne Prize for seventy-
 167 six years. First awarded on 14 March 1950, the prize honours individuals who have made
 168 outstanding contributions to peace, unity, and cooperation in Europe [14]. Its creation coincided
 169 with a formative decade of European integration. The decade witnessed the establishment of
 170 the European Coal and Steel Community (ECSC) through the Treaty of Paris (1951; in force
 171 from 1952) and, later, of the European Economic Community (EEC) and the European Atomic
 172 Energy Community (EURATOM) through the Treaties of Rome (1957; in force from 1958). Less
 173 frequently noted is that the combined territory of the six states that founded the ECSC, the EEC,
 174 and EURATOM overlaps substantially with the core regions of the Carolingian Empire. This
 175 overlap is noteworthy because Aachen is closely associated with Charlemagne, who established
 176 a palace complex (Pfalz) there and made the city a center of Carolingian rule. Aachen therefore
 177 occupies a distinctive place in both the symbolic landscape of post-war European integration
 178 and the historical geography of the Carolingian world.

179 The Carolingian connection is also visible at RWTH Aachen University. It is reflected, among
 180 other things, in the acronym C.A.R.L., the name of a representative university building that
 181 served as the venue for CoRDI 2025 [9]. Moreover, the local host of CoRDI 2025, Robert
 182 Schmitt of NFDI4ING, referred explicitly to the Carolingians in his opening remarks of the
 183 CoRDI. Against this backdrop of Carolingian references, Évariste was reminded of a familiar
 184 narrative convention in comics: the use of an introductory map, as found, for example, on the
 185 front endpapers of *Astérix* albums [15]. This convention inspired the idea of opening the FAIR
 186 comic with a map of the Carolingian Empire.

187 The first three pages of the FAIR comic are dedicated to establishing context and continuity

(cf. chapter five in [5]) within the broader geographical and cultural landscape of NFDI4ING. They are also intended to entertain readers and draw them toward the more technical aspects of research data infrastructure. To this end, a small number of fictional elements were introduced, including reconstructions of what Charlemagne and his descendants might have said and done up to the partition of the empire under the Treaty of Verdun in 843 AD. In designing and drawing the first three pages, Alfred used expertise he had acquired in the course of an earlier project [16]. The map of the Carolingian Empire, specifically, is based on a historical map distributed through Alamy [17]. On a more technical note, most of the details of the historical map were omitted in order to emphasize selected features, while a few others were added. Among these is the empire’s eastern frontier, which is marked on the map by the added detail *limes sorabicus*. The territories beyond this frontier were subject to varying degrees of Carolingian rule and correspond broadly to the toponomastic boundary between regions characterized by Germanic and Slavic place-name strata [18]. Another addition is the city of Darmstadt, home to Technische Universität Darmstadt, applicant institution of NFDI4ING during its second funding period [19].

As soon as the FAIR comic shifts from the Carolingian past to present-day Aachen and C.A.R.L., the staged setting of the meeting between Alfred and Évariste, the colour palette switches to indicate a different register: conversation. The ensuing dialogue depicts discussions that took place over the course of almost five months of weekly meetings involving Alfred and either Évariste alone or Évariste and Tobias together. The FAIR comic organizes these discussions thematically and condenses them into nine pages.

3.3 Manuscript Preparation and Translation

The process of outlining, designing, and refining the FAIR comic was conducted in German. Accordingly, the first draft was written in German. As we approached publication of the FAIR comic in *ing.grid*, however, it had to be translated into English. At the same time, we considered making the FAIR comic available in different languages important for its accessibility. One way of doing so would have been for Alfred to redraw all text in the FAIR comic for each additional language. This would have resulted in lengthy review cycles and placed a considerable additional workload on Alfred.

Tobias instead developed a more automated approach to producing multilingual versions. Alfred created a blank PDF version of the FAIR comic containing the interlocutors, maps, and all other graphical elements, but no text. This version served as the basis for a L^AT_EX document in which each blank comic page was used as the background of a full-page TikZ image [20]. TikZ is a recursive acronym for *TikZ ist kein Zeichenprogramm*, German for “TikZ is not a drawing programme”. Alfred’s original German text was then superimposed on the pages using TikZ nodes (see the [German version in the appendix](#)). These nodes can be positioned freely and styled individually, allowing precise control over fonts, font sizes, and special text arrangements, including curved or rotated text. Manual line breaks can likewise be specified in the language file, allowing the text layout to be adjusted to achieve the visual balance intended by the artist. One language-dependent element, however, could not be handled as a textual overlay. Alfred also had to remove the drawing of three-dimensional toy bricks forming the letters “FDI”, standing for the German *Forschungsdateninfrastruktur*, from the first page. Here, a language-dependent version of the image was required. To address this, different language versions of the image

were generated using artificial intelligence (AI), namely OpenAI’s ChatGPT 5.5. The Artificial Intelligence (AI) was instructed to preserve the style of the original image while replacing the hand-sketched “FDI” with “RDI”.

Similarly, the language file can be translated automatically using AI. To test this concept, we had AI translate the language file into Latin and J. R. R. Tolkien’s Elvish, thereby generating corresponding versions of the FAIR comic. Both versions have a dextrograde (left-to-right) writing direction. The Elvish version, rendered in the Tengwar script, further demonstrates that non-Latin scripts can be accommodated. Sinistrograde (right-to-left) and boustrophedon (alternating writing direction from line to line) writing have not yet been implemented or tested.

The source code is available on [GitHub](#)¹ [21]. Every language CODE is kept in its own folder `language_files/CODE`, which holds the text file `CODE.tex`, the corresponding folder `pages_CODE` and all language dependent images. Here, CODE is the lowercase three-letter identifier from ISO 639 Set 3 (formerly ISO 639-3) [22], as used by the Zenodo language vocabulary. The Draconic version uses the project-local placeholder `vss`, derived from the Draconic name *Vs’shtak*; this identifier is neither registered in ISO 639 nor in Zenodo. If additional fonts are needed, they are contained in the subfolder `fonts_CODE`. As a wrapper, the file `comic_files/pages_functions.tex` is processing these folders. Every `language_files/CODE/CODE.tex` contains all text lines ordered by their appearance on the pages in the corresponding language, which allows for a translation of texts independently from their appearance in the document. Also, manual line breaks can be added here as needed to account for the styling and weighting of texts within speech bubbles. The `pages_CODE` files handle the positioning and scaling of texts for each language individually, which accounts for different lengths of the same contents in different languages. The language is chosen in the `\comic{CODE}` command. It outputs all comic pages in the selected language one after another, e.g. `\comic{eng}` generates the English version of the comic. Since each call is self-contained, several language versions can also be printed one after another within the same document. The primary comic language and the language-specific title, subtitle, and short title are selected in `main.tex` with `\newcommand{\FDILanguage}{eng}`. The article body and the remaining metadata are maintained separately. As of writing, the original German version and this translated English version exist along the Latin, Elvish, and Draconic versions.

4 Discussion

4.1 Dialogue as a Documentary and Didactic Form

The conversational format of the FAIR comic has a philosophical pedigree reaching back to the dialogues of Plato (*circa* 424/3–348/7 BC [23]). Parallels include the use of historical interlocutors [23] and a characteristic division of labour: Socrates asks questions, while his interlocutors formulate and defend answers. Like Socrates, Alfred frequently advances the discussion through questions. One of the most arresting things about the Platonic dialogues is that most of them have survived for more than two millennia and continue to be widely read and discussed today. We contend that such an unusually long period of transmission should be taken

1. <https://github.com/tobias-hamann/research-comic-vol1>

as an indication of the longevity not only of their content but also of their form. References to Plato continue to appear in contemporary culture and education. Examples include Jay-Z, Kanye West, Frank Ocean, and The-Dream's song "No Church in the Wild", which invokes the *Euthyphron* dilemma [24], [25]; Helme Heine's children's book *Tante Nudel, Onkel Ruhe und Herr Schlau*, which adapts the *Politeia* [26]; Höfner and Süßbier's didactic comic *Das verrückte Mathe-Comic-Buch*, which includes a re-telling of the *Menon* [27]; as well as James Orr's lecture series *Plato* [28]. This continued presence suggests not only the enduring relevance of the subject matter, but also the effectiveness and remarkable adaptability of the dialogical format through which that subject matter is explored. As Orr points out in the opening lecture of his series, "once you know all of the sides, you understand your own" [28, at 55:09]. The FAIR comic similarly relies on dialogue as a means of exploring and communicating complex, albeit technological, matters. We hope that readers interested in research data infrastructure will be profitably entertained by the result.

We have introduced the FAIR comic by reference to the documentary practices of ethnographic research [3], [4], and discourse and conversation analysis [5], [6]. More specifically, we have situated comic culture within a continuum of practices concerned with documenting deixis and embodied speech (see 1). We have argued that, as a graphic medium, comics are particularly well suited to documenting multimodal interaction in situated research environments. In this sense, the FAIR comic may itself be understood as a documentary form of speech. If we have now reached back to the dialogues of Plato, it is still in an attempt to reflect on the structure of the FAIR comic and, more generally, on what helps readers understand the spoken word [29].

4.2 The FAIR Comic between Research and Science Comics

We conclude furthermore our discussion by comparing the FAIR comic to related forms of comic culture, such as comics about researchers in research institutions (see, for example, Jorge Cham's *PHD Comics* or Eppendorf's *Lab Life*; [30], [31]), as well as science communication comics that explain fields of research (see, for example, Larry Gonick's *Cartoon Guide* series or Matteo Farinella's *Neurocomic*; [32], [33], [34]). The FAIR comic sits comfortably between these two broader lineages and follows a journalistic ethos. What we seek to achieve is not adequately captured by saying that the FAIR comic is about research practice, nor even that it is researched, but rather that it is explicit about the process of learning—or coming to understand. To put our claim in direct speech: "This is what we discussed while trying to establish a mutual understanding of research data infrastructure."

Von Karl dem Großen bis zu Karl dem Kleinen.
 Von der Karolingischen Minuskel über das Mittelreich bis zu  NFDI4ING.*
 Forschungsdateninfrastruktur ist und bleibt eine spannende Sache.



* NFDI4ING ist das Konsortium für die Ingenieurwissenschaften in der Nationalen Forschungsdateninfrastruktur (NFDI)

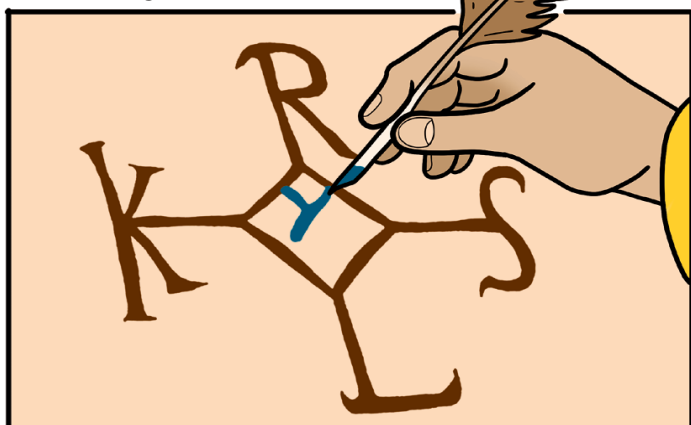
303

Im 8./9. Jahrhundert begann Karl der Große mit seiner zentralisierten Schriftkultur eine nachhaltige Wissensinfrastruktur aufzubauen. Er stützte sich dabei auf ein Netzwerk von Skriptoren in Klöstern und an seinem Hof in Aachen.



304

Karl selbst konnte nicht schreiben. Beim Unterschreiben von Dokumenten fügte er dem Karolus-Monogramm lediglich den sogenannten Vollziehungsstrich hinzu.



Karl förderte aber Bildung durch das Diktieren, Kopieren und die Verbreitung von Dokumenten durch Gelehrte.

305



306

Um die Lesbarkeit und Einheitlichkeit von Dokumenten in seinem riesigen Reich zu gewährleisten, ließ er die "Karolingische Minuskel" entwickeln.

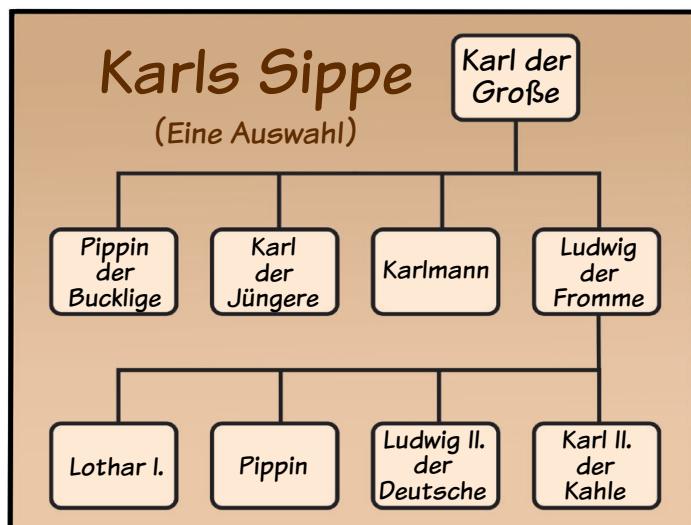


307



308

Karls Söhne und Enkel setzten diese analoge Forschungsinfrastruktur auf Basis von Pergament und Tinte fort.



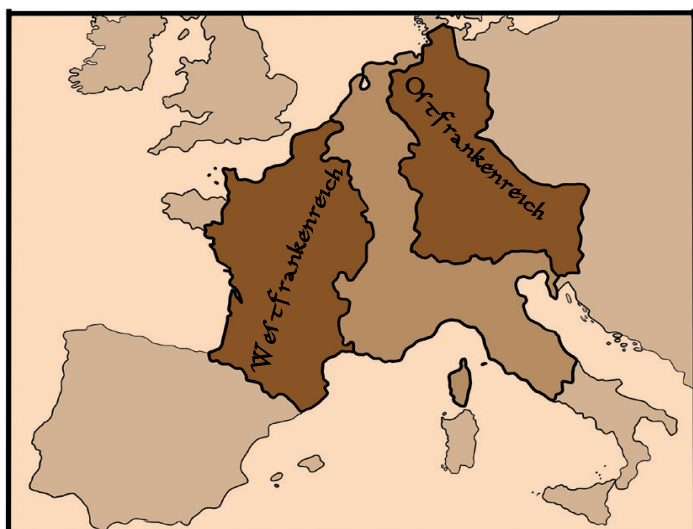
309

Einer seiner Enkel, Lothar I., war im Jahr 843 maßgeblich an dem Vertrag von Verdun beteiligt.



310

Dieser Vertrag gilt als Geburtsstunde der Vorläufer von Frankreich und Deutschland (Westfrankenreich und Ostfrankenreich).



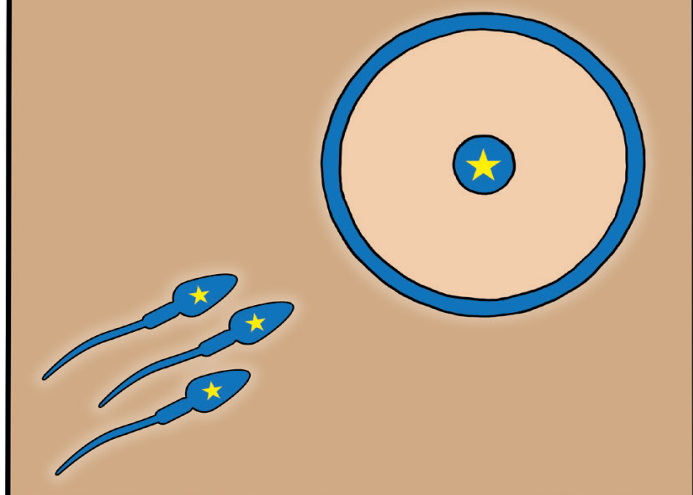
311

Lothar selbst erhielt das Mittelreich. Es erstreckte sich als langer, schmaler Korridor von der Nordsee bis zum Mittelmeer.



312

Dieses Mittelreich wurde zur Keimzelle der europäischen Integration.



313

Viele Institutionen der Europäischen Union sitzen heute entlang dieses ehemaligen Korridors (Brüssel, Luxemburg, Straßburg, Rom).



314

So kommt es, dass die Comiciade im Technologiezentrum am Europaplatz in Aachen ihre Tore für Comicfans aus aller Welt öffnet.



316



315

Auch im C.A.R.L. der RWTH Aachen laufen hierzu bereits die Vorbereitungen.



317



318



319

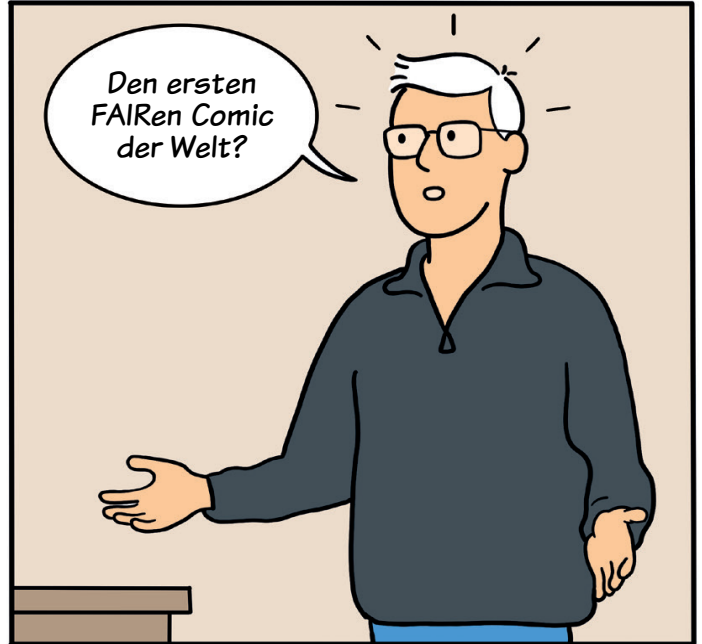


320



Dein Comic über Deine Zeit am Lehrstuhl für Methodik der künstlichen Intelligenz hat mir sehr gut gefallen. Was hältst du davon, jetzt den ersten FAIRen Comic der Welt zu machen?

321



Den ersten FAIRen Comic der Welt?

322



Ich meine FAIR im Sinne der Forschungsdateninfrastruktur (FDI).

Die FAIR-Prinzipien sind hierbei: Findable, Accessible, Interoperable und Re-usable.*

323



Das musst Du mir näher erklären, Évariste.

324



Ganz einfach. Du hast auf der Rückseite Deines Comics eine ISBN-Nummer angegeben, damit er im Buchhandel leichter zu finden ist.

325



*Auffindbar, zugänglich, interoperabel und nachnutzbar

326

Man benötigt eine ISBN-Nummer, um in das Verzeichnis lieferbarer Bücher (VLB) aufgenommen zu werden. Wenn man dort nicht gelistet ist, ist es sehr schwer, sein Buch in die Läden zu bekommen.



327

Das VLB ist die zentrale, unabhängige Datenbank und das Recherchetool für den deutschsprachigen Buchhandel.



328

Interessant. Wir Wissenschaftler benutzen anstatt einer ISBN-Nummer eine DOI (Digital Object Identifier).



329

Sie ist ein eindeutiger, dauerhafter Identifikator für Objekte aller Art. Ein Objekt kann mehrere DOIs haben, aber eine DOI zeigt immer nur auf ein Objekt.

DOI-Auflöser

Repository

<https://doi.org/10.5281/zenodo.19468332>

Präfix

Oder als
QR-Code

Objekt

330



Ein Partner von NFDI4ING, der DataCite-Verein, vergibt DOIs.*



331

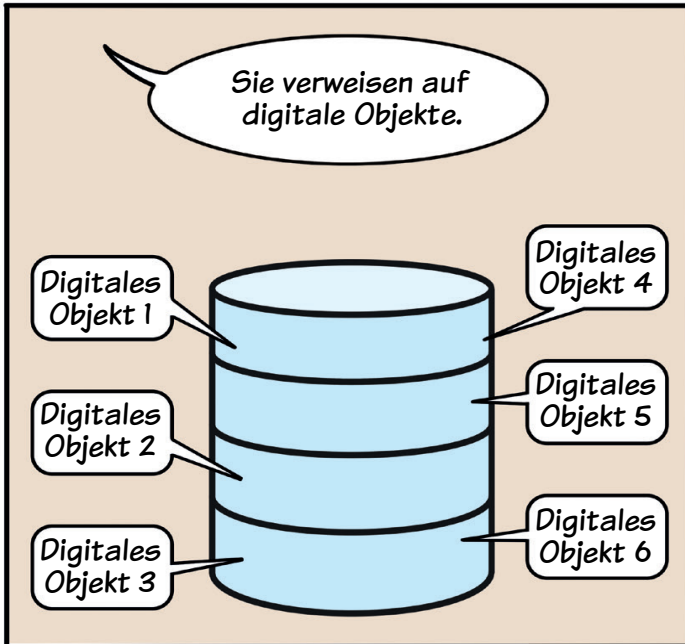
DOIs sind bei wissenschaftlichen Arbeiten der Standard für die präzise Quellenangabe.

Und wie genau funktionieren sie?



*DataCite ist eine Registrierungsagentur für DOIs

332



333



334



335



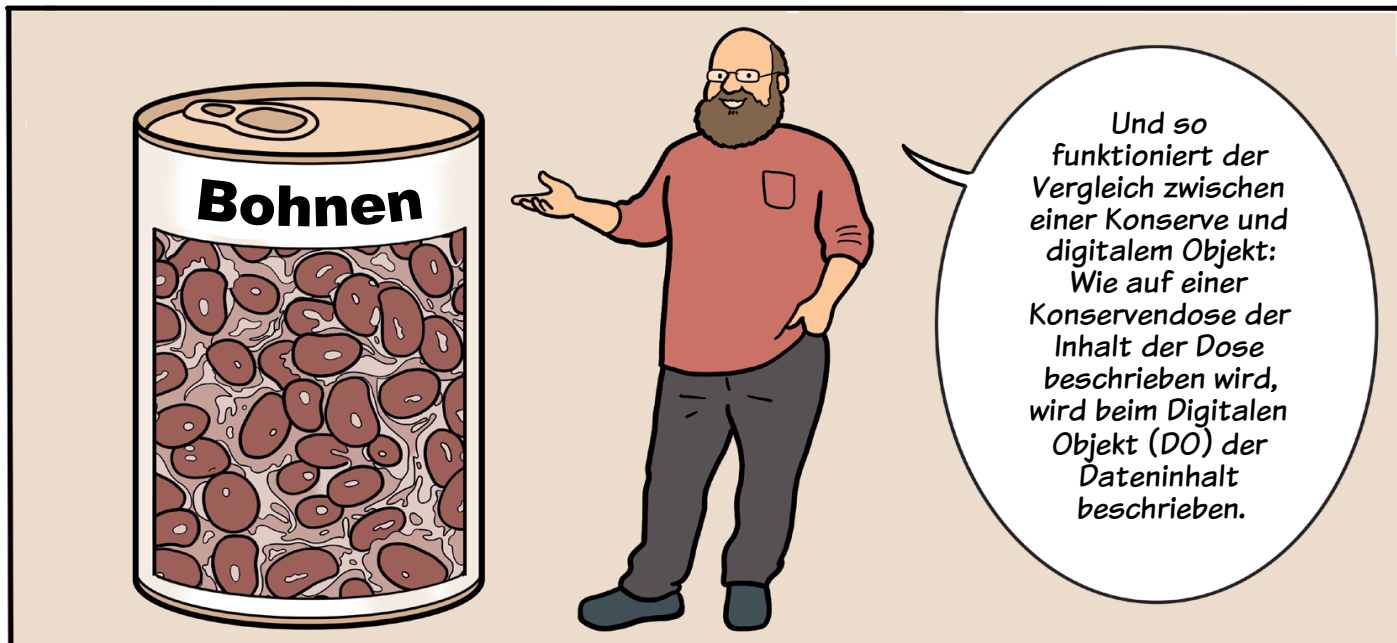
336



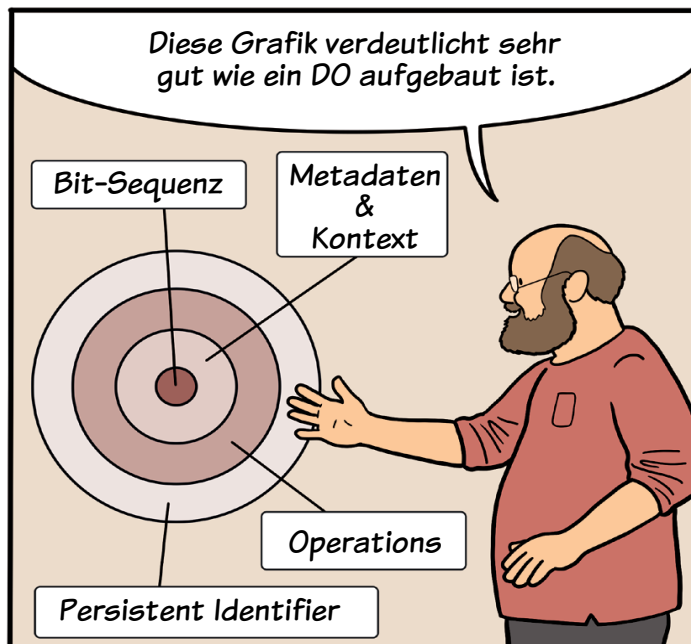
337



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348



*URN = Uniform Resource Name, Handle = Handle System; ORCID = für Open Researcher and Contributor ID

349

In der FDI beschreibt der Begriff "Alphabetisieren" den Prozess des Erwerbs grundlegender Fähigkeiten und Kenntnisse, die notwendig sind, um Forschungsdaten lesen und schreiben zu können.



350

Wenn sie niemand lesen kann, nützt auch das beste Archiv nichts und die Daten bleiben stumm.



Das wusste auch schon Karl der Große. Er hat erkannt, dass Infrastruktur ohne Bildung nutzlos ist.

351

Die
Datensprache
muss man lesen
lernen.



352

Karl der Große legte großen Wert darauf, dass Kleriker und Verwaltungsbeamte lesen und schreiben können. Auch, damit sie korrekt abschreiben. Fehlerhafte Datenübertragung war schon damals ein Problem.



353

Sozusagen
Copy-Paste-Fehler im
9. Jahrhundert.

HA HA HA!
HA HA HA!
HA HA HA!

Das nenne
ich doch mal
eine gelungene
Interoperabilität von
Vergangenheit und
Gegenwart!



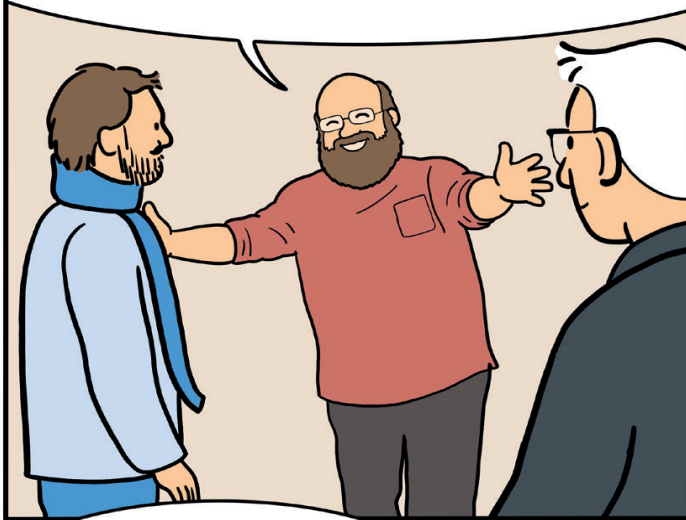
354

Was gibt
es denn hier zu
lachen?



355

Hallo Tobias. Wir sind gerade dabei, einen FAIREN Comic zu gestalten, und das ist gerade sehr unterhaltsam.



356

Das ist ja spannend. Ich habe gestern mein neuestes Paper bei ing.grid eingereicht. Das könntet ihr doch mit diesem Comic auch machen.



357

ing.grid ist ein Diamond-Open-Access-Journal der NFDI4ING.

Dort gibt es auch einen Preprint-Server.



358

Das ist eine tolle Idee, Tobias. Wenn wir dort einreichen, wäre der Comic schon auf dem Preprintserver sichtbar und für alle zugänglich.



359

Genau! Ein Community-Review wäre dort der erste Schritt. Das heißt, der Comic könnte dann bereits kommentiert und bewertet werden.



360

Super! Wo soll ich meinen Vollziehungsstrich hinsetzen?



361



362



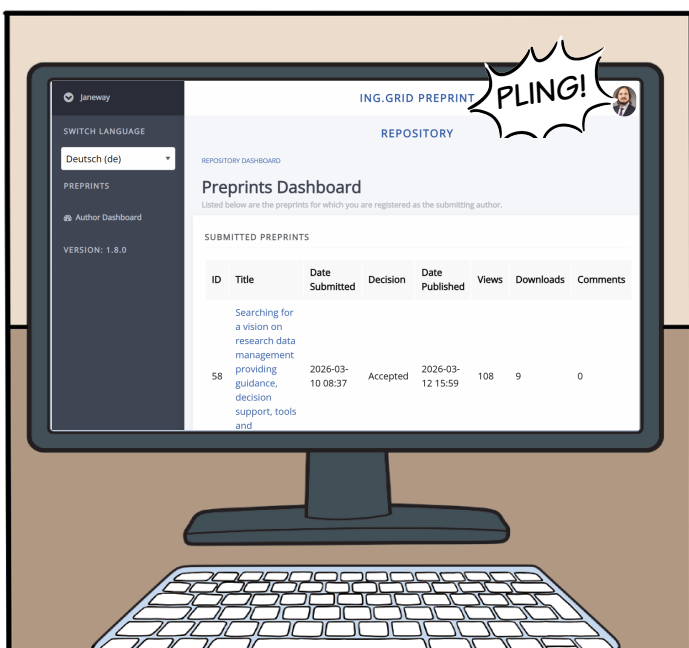
363



364



365



366



* PDF = Portable Document Format

366 B The comic in Vs'shtak (Draconic, planned)

367 *The rendered Draconic edition is being prepared and will be added in a future version.*

368 C Acknowledgements

369 The authors would like to thank the Federal Government and the Heads of Government of the
 370 Länder, as well as the Joint Science Conference (GWK), for their funding and support within the
 371 framework of the NFDI4ING consortium. Funded by the German Research Foundation (DFG),
 372 project number 442146713 [19]. The authors would further like to thank Lukas C. Bossert,
 373 Zeynep Dirier, Thomas Eifert, Marius Politze, Heinrich Schlange-Schönigen, Ines Scholten-
 374 Strauch, Benedikt Schmitz, and Annett Schwarz for discussions and helpful comments during
 375 the course of the project.

376 D Roles and contributions

377 **Alfred Neuwald:** Conceptualization, Methodology, Writing, Visualisation

378 **Tobias Hamann:** Conceptualization, Methodology, Writing, Software

379 **Évariste Demandt:** Conceptualization, Writing

380 References

- 381 [1] J. S. Bak and C. Craig, *Comics Journalism, BD Reportage, Documentary Manga: Graphic*
 382 *Literary Journalism around the World*, Nancy, May 2026. [Online]. Available: https://mangix.event.univ-lorraine.fr/data/pages/Summer_School_2026_course_program_3.pdf
 383
 384
 385 [2] A. Neuwald (Neufred), *DEUS AIX MACHINA*. Aachen: Granus Verlag, 2026, ISBN:
 386 978-3-9822083-4-3.
 387 [3] B. Latour and S. Woolgar, *Laboratory Life: The Social Construction of Scientific Facts*,
 388 1st ed. Beverly Hills, CA: Sage Publications, 1979.
 389 [4] B. Latour and S. Woolgar, *Laboratory Life: The Construction of Scientific Facts*, 2nd ed.
 390 Princeton, NJ: Princeton University Press, 1986.
 391 [5] D. Edwards and N. Mercer, *Common Knowledge: The Development of Understanding in*
 392 *the Classroom*. 1st ed. London: Methuen & Co. Ltd., 1987.
 393 [6] I. Hörmeyer, *Der Einsatz von Körper und Maschine in der unterstützten Kommunikation:*
 394 *eine konversationsanalytische Untersuchungen* (Empirische Kommunikationsforschung
 395 im Gesundheitswesen Bd. 4). Mannheim: Verlag für Gesprächsforschung, 2015, ISBN:
 396 978-3-936656-67-1.
 397 [7] J. S. Bak and C. Craig, *Mangix: Tohoku–Lorraine Research Project in Comics/Manga*
 398 *Journalism*, Nancy / Sendai, Sep. 2024. [Online]. Available: [https://idea.univ-lorr](https://idea.univ-lorraine.fr/sites/default/files/users/user844/Mangix%20Tohoku-Lorraine%20research%20project%20in%20comics-manga%20journalism.pdf)
 399 [aine.fr/sites/default/files/users/user844/Mangix%20Tohoku-Lorr](https://idea.univ-lorraine.fr/sites/default/files/users/user844/Mangix%20Tohoku-Lorraine%20research%20project%20in%20comics-manga%20journalism.pdf)
 400 [aine%20research%20project%20in%20comics-manga%20journalism.pdf](https://idea.univ-lorraine.fr/sites/default/files/users/user844/Mangix%20Tohoku-Lorraine%20research%20project%20in%20comics-manga%20journalism.pdf)

- [8] L. Simons, “Traduction commentée de la bande dessinée aixoise Karl der Kleine: Die Printe des Teufels d’Alfred Neuwald,” M.S. thesis, Université de Liège, Liège, 2021. [Online]. Available: <http://hdl.handle.net/2268.2/12918>
- [9] N. e.V., *CoRDI 2025 | NFDI*. Accessed: Jun. 12, 2026. [Online]. Available: <https://www.nfdi.de/cordi-2025/>
- [10] A. Spiegelman, *Maus 1: A Survivor’s Tale: My Father Bleeds History*. New York: Pantheon Books, 1986.
- [11] M. Satrapi, *Persepolis: The Story of a Childhood*. New York: Pantheon Books, 2003.
- [12] G. Delisle, *Pyongyang*. Paris: L’Association, 2003.
- [13] Landesinitiative NFDI Der Digitalen Hochschule NRW and AG FDM Awareness, *Nachnutzbare Awarenessmaterialien für Forschungsdatenmanagement (FDM)*, Mar. 2019. DOI: [10.5281/ZENODO.2599866](https://doi.org/10.5281/ZENODO.2599866) Accessed: Jun. 10, 2026. [Online]. Available: <https://zenodo.org/record/2599866>
- [14] S. I. K. zu Aachen, *Start - Der Internationale Karlspreis zu Aachen*. Accessed: Jun. 10, 2026. [Online]. Available: <https://www.karlspreis.de/en/start>
- [15] R. Goscinny and A. Uderzo, *Astérix le Gaulois*. Paris: Dargaud, 1961.
- [16] A. Neuwald (Neufred), *Karl der Kleine bei den Karolingern: Der Comic zum Karlsjahr 2014*. Aachen: Aachener Karnevalsverein, 2014.
- [17] *Europe843 Stockfotografie - Alamy*, Abingdon, Oxfordshire. Accessed: Jun. 10, 2026. [Online]. Available: <https://www.alamy.de/stockfoto-europe843-139813809.html>
- [18] M. Jelić, “Das Ethnonym Srbi und seine onymischen und deonymischen Ableitungen in der Slavia,” M.S. thesis, Ruhr-Universität Bochum, Bochum, 2019. [Online]. Available: https://hbz-ubo.primo.exlibrisgroup.com/permalink/49HBZ_UB0/1roc4t0/alma991019690469706471
- [19] *DFG - GEPRIS - NFDI4ING – Nationale Forschungsdateninfrastruktur für die Ingenieurwissenschaften*. Accessed: Jun. 18, 2026. [Online]. Available: <https://gepris.dfg.de/gepris/projekt/442146713>
- [20] T. Tantau, “Graph drawing in TikZ,” in *Graph Drawing* (Lecture Notes in Computer Science), W. Didimo and M. Patrignani, Eds., Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, vol. 7704, pp. 517–528, ISBN: 978-3-642-36762-5. DOI: [10.1007/978-3-642-36763-2_46](https://doi.org/10.1007/978-3-642-36763-2_46) [Online]. Available: http://link.springer.com/10.1007/978-3-642-36763-2_46
- [21] T. Hamann, *A FAIR Comic about Research Data Infrastructure — LaTeX sources*, version 0.1.0, Aug. 2026. DOI: [10.5281/zenodo.21921316](https://doi.org/10.5281/zenodo.21921316) [Online]. Available: <https://github.com/tobias-hamann/research-comic-vol1>
- [22] International Organization for Standardization, *ISO 639:2023: Code for Individual Languages and Language Groups*, Geneva, Nov. 2023. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.iso.org/standard/74575.html>
- [23] N. Debra, *The People of Plato: A Prosopography of Plato and Other Socratics*. Indianapolis, IN: Hackett Publishing, 2002.

- 442 [24] Jay-Z, K. West, F. Ocean, and The-Dream, *No Church in the Wild*, New York, 2011.
- 443 [25] Jay-Z & Kanye West - *No Church In The Wild ft. Frank Ocean, The-Dream*, May 2012.
- 444 Accessed: Jun. 18, 2026. [Online]. Available: <https://www.youtube.com/watch?v=FJt7gNi3Nr4>
- 445
- 446 [26] H. Heine, *Tante Nudel, Onkel Ruhe und Herr Schlau: ein Bilderbuch*. Köln: Middelhaue,
- 447 1979, ISBN: 978-3-7876-9920-9.
- 448 [27] G. Höfner and S. Süßbier, *Das verrückte Mathe-Comic-Buch*. Heidelberg: Spektrum
- 449 Akademischer Verlag, 2012, ISBN: 978-3-8274-2629-1. DOI: [10.1007/978-3-8274-2](https://doi.org/10.1007/978-3-8274-2629-1)
- 450 [629-1](https://doi.org/10.1007/978-3-8274-2629-1) Accessed: Jun. 22, 2026. [Online]. Available: [http://link.springer.com/10](http://link.springer.com/10.1007/978-3-8274-2629-1)
- 451 [.1007/978-3-8274-2629-1](http://link.springer.com/10.1007/978-3-8274-2629-1)
- 452 [28] P. Academy and J. Orr, *Dr. James Orr | Plato: The Dawn of Thought | Lecture 1 (Official)*,
- 453 May 2026. Accessed: Jun. 10, 2026. [Online]. Available: [https://www.youtube.com](https://www.youtube.com/watch?v=QwDoENWBId0)
- 454 [/watch?v=QwDoENWBId0](https://www.youtube.com/watch?v=QwDoENWBId0)
- 455 [29] W. J. Ong, *Orality and Literacy: The Technologizing of the Word*, 1st ed. Methuen & Co.
- 456 Ltd., 1982.
- 457 [30] J. Cham, *PHD Comics: NEW BOOK! Out of Your Mind!* 1997. Accessed: Jun. 22, 2026.
- 458 [Online]. Available: <https://phdcomics.com/>
- 459 [31] *Lab Life - Eppendorf Netherlands*. Accessed: Jun. 22, 2026. [Online]. Available: [https:](https://www.eppendorf.com/nl-en/beyond-science/lab-life/)
- 460 [/www.eppendorf.com/nl-en/beyond-science/lab-life/](https://www.eppendorf.com/nl-en/beyond-science/lab-life/)
- 461 [32] L. Gonick and M. Wheelis, *The Cartoon Guide to Genetics*. New York: Harper & Row,
- 462 1983.
- 463 [33] L. Gonick and D. Smith, *The Cartoon Guide to Computer Science*. New York: William
- 464 Morrow, 1983.
- 465 [34] M. Farinelli and H. Roš, *Neurocomic*. London: Nobrow Press, 2013.